



Zahir Khan
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Ramish Masood
2020-CE-086

Project# 20

Augmented Reality Based Computer Fundamentals Course

Internal: Miss Mehjabeen Tahir

Adil Aljaz
2020-CE-095

Qasim
2020-CE-113



Project Objectives

In many educational environments, students do not have sufficient access to computer components to gain hands-on experience. Traditional teaching methods based on static images or diagrams to illustrate hardware components do not offer interactive experience for students.

The objective of our project is to overcome this problem and enhance learning process by showing the computer hardware and its devices using AR technology in a way that students can visualize the computer hardware & devices easily in 3D from different angles on the App using their mobile phones.

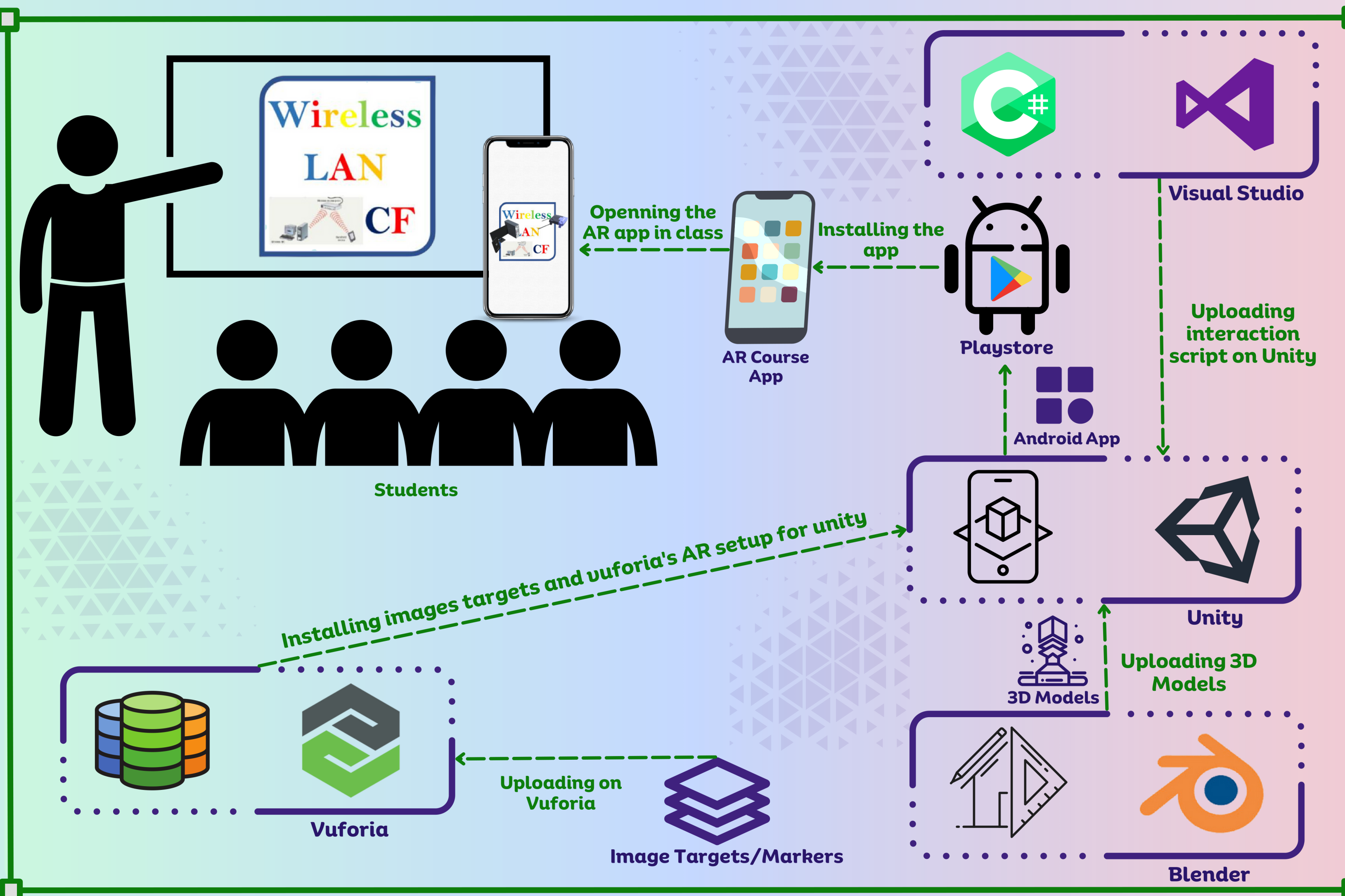
Project Features

3D Visualization of Computer Components and Devices: Students can visualize computer components and devices using AR 3D models

Interactive 3D Models: Students can interact with the 3D models by rotating them in all directions using touch gestures on their mobile devices.

Offline Access: Students will be able to access the AR content offline after downloading the android app.

Support for Different Devices: The android app is optimized to work on a wide range of Android devices. (Android 8.0 and above)

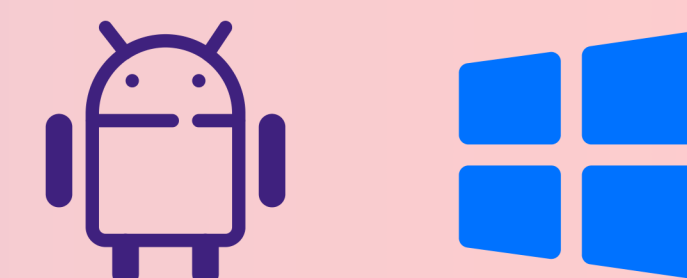


Brief Explanation

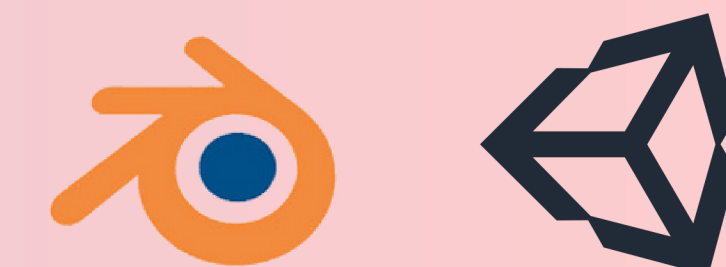
Our Project is an application that will help students to visualize the computer hardware and components in 3D for better learning experience & understanding of computer components. It is built on Unity Game Engine which uses Vuforia AR SDK technology. The App consists of different modules. 3D models of hardware components are designed on Blender. The detection of the hardware components on the screen is done by using Image Targets stored in Vuforia DB. C# scripts are also used for rotation of 3D objects. The UI of the App will be designed using UI toolkit on Unity Editor. All these different modules of our App are then integrated into a single standalone App which will be uploaded to Google Play Store, from which users can download the App and use it for learning purposes.

Tools

Operating System



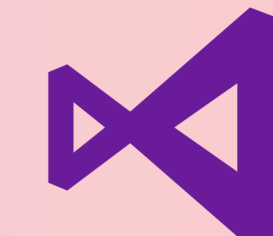
Designing Tools



Database and AR Development kit



IDE



Project Scope

The scope of our Project encompasses the development of an interactive, engaging, and immersive learning platform to educate students on the fundamental concepts of computer hardware. The App will be a valuable resource for users of all ages and skill levels who want to learn about computer hardware & its components.